KRISH JOSHI

AI Evaluation Engineer | LLM Benchmark Design • Python • C/C++ • GPU Kernels (OpenCL)

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PROFESSIONAL SUMMARY

CS undergraduate (B.Tech, 2028) who builds challenging, auto-graded benchmark tasks evaluating frontier AI coding agents. Delivered 10 containerized Harbor-format environments spanning GPU kernels, discrete algorithms, storage recovery, and distributed systems during a completed 3-month contract for Handshake. Strong in Python, PyTest, Docker, Linux, and low-level C/OpenCL with hands-on experience designing deterministic test harnesses and calibrating model difficulty via empirical pass@k trial regimes.

WORK EXPERIENCE

AI Benchmark Developer (Independent Contractor) **July 2026 – September 2026**

Handshake — AI Evaluation Program (Project Dynamo) Remote (Contract Completed)

- Authored 10 terminal-based engineering benchmarks in Docker (Harbor format) evaluating frontier coding models across low-level GPU compute, Linux storage recovery, and distributed DAG execution.
- Engineered deterministic PyTest graders (CTRF schema, SHA-256 assertions, IEEE-754 float tolerances) where reference solutions passed 100% while frontier models failed on ≥60% of initial runs.
- Calibrated task difficulty using pass@2 and pass@5 trial regimes against frontier models; iterated prompt constraints until tasks reliably stumped models attempting naive or memorized heuristics.
- Hardened sandboxed evaluation runtimes with strict path-traversal assertions (assert_safe_path) and memory/CPU quotas, preventing environment tampering and prompt-injection bypasses.

SELECTED BENCHMARK TASKS (PROJECT DYNAMO)

W4A16 Quantized GEMM GPU Kernel Benchmark **OpenCL C, C/C++, Python, PyTest, SIMT Concurrency**

- Engineered a high-throughput matrix multiplication kernel executing tiled dot products with asymmetric 4-bit packed weights and FP32 accumulation; eliminated workgroup races using local memory barrier fences.
- Automated verification across non-power-of-two matrix shapes against FP64 reference baselines; caused naive agent implementations to fail on memory bounds and ragged tensor alignments.

Distributed Execution Graph & DAG State Reconciliation **Python, Graph Algorithms, SQLite, PyTest, JSONL**

- Architected an evaluation task requiring models to parse 3,500+ line cluster execution logs and reconstruct valid state DAGs across branching rollbacks and node failures.
- Graded solutions on recursive cycle detection, tree pruning, and deterministic tie-breaking with ±1.0s wall-clock accuracy, eliminating superficial string-matching shortcuts.

Octahedral Rotation Group 3D Constraint Solver **Python, NumPy, Discrete Mathematics, CSP Backtracking**

- Designed a combinatorial benchmark modeling 3D orientation constraints over the 24-element octahedral group (subgroup of SO(3)) using non-commutative composition paths.
- Implemented depth-first search with forward checking and matrix trace parity verification; benchmarked model ability to systematically prune search branches without timing out.

TECHNICAL SKILLS

- **AI Evaluation & Benchmarking:** Benchmark Design (Harbor format), pass@k Calibration, Deterministic Test Harnesses, Sandbox Isolation, Agent Transcript Review
- **Programming Languages:** Python (PyTest, NumPy), C/C++, OpenCL C, Bash Scripting, SQL (SQLite)
- **Systems & Testing:** Docker Containerization, Linux CLI & Utilities, Git & GitHub, CTRF Schema Reporting, IEEE-754 Precision Testing, SHA-256 Checksums
- **Core Computer Science:** Data Structures & Algorithms (DSA), Concurrency (SIMT Barrier Fences), Graph Algorithms (DAGs), Discrete Mathematics

EDUCATION & CERTIFICATIONS

Bachelor of Technology (B.Tech), Computer Science & Engineering **July 2024 – June 2028 (Expected)**

Institute of Advanced Research (IAR) — Gandhinagar, India

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks
- **Certifications:** Google Cybersecurity Professional Certificate (Coursera / Google) | Introduction to Networking (NVIDIA)